



OxDolphin

The Next BIG Thing in Security.

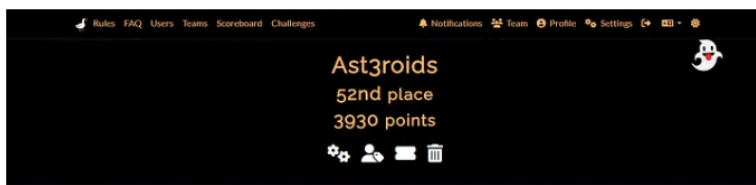
Steganography + Ctf + Ctf Writeup + Forensics + Ctf Walkthrough +

V1t CTF 2025- "67"

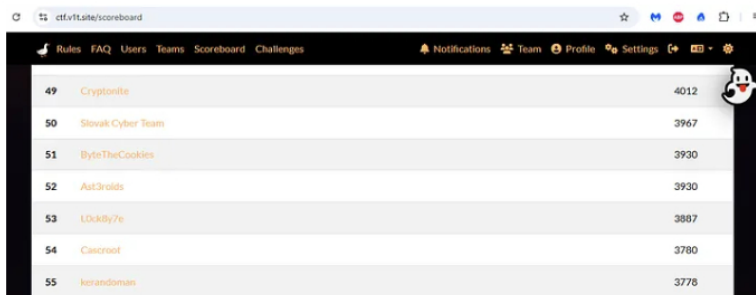


OxDolphin 4 min read · Nov 4, 2025

10 2



Hello guys, this is Mousa AKA OxDolphin. I took part with my amazing team Ast3roids in v1t CTF 2025, proudly achieving rank #52/ 1237 teams in my first official CTF after my comeback. I am very happy that we were in the top 5%. I want to share writeups and analysis for some good challenges in MISC and FORENSICS.



67

<https://ctf.v1t.site/challenges#67-69>

This was one hell of a stegno challenge tbh. Huge shoutout to my colleague T4MER for rocking it out with me. < 3

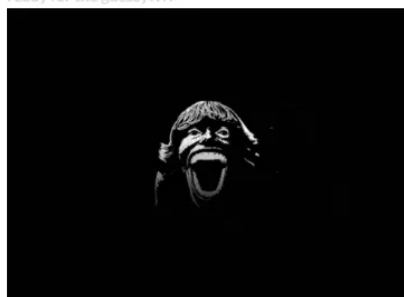
67

460

Jaze

0 (0% liked) 1

ready for the guessy????



[challenge.zip](#)

Challenge

first, lets take a look on the zip file.

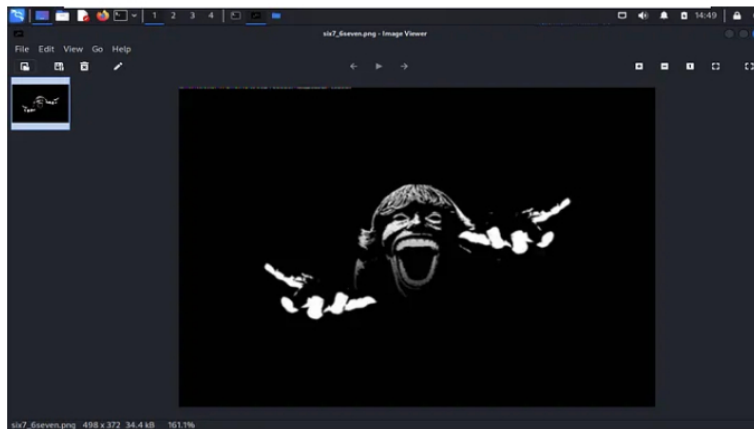
```
(kali@kali)-[~/Downloads/forensics_ctflearn/zz]
$ strings challenge.zip
=;_[
six7_6seven.pngUT
```

strings

```
:3+|
57`Of
Vy2o
d`x
H6trd
b9n!
OT>u
gl^:7x
ed32a
Ael(\
[zH'X
T@u@
I)IVH
=;_[
six7_6seven.pngUT
u AlrEady know the ruleS | (36-4) + (36-4) = 67 - 3.
```

1st hint

Mmm. Ok, remeber it for now. Lets see the photo.



ok. now we anazlye it using our normal workflow. take care that there are some colorful pixels which can mean that there wa ssome LSB manipulation. so maybe zsteg?

```
(kali@kali)-[~/Downloads/forensics_ctflearn/zz]
$ xdg-open six7_6seven.png

(kali@kali)-[~/Downloads/forensics_ctflearn/zz]
$ zsteg six7_6seven.png
imagedata
.. text: ">?CDB=6%%"
b2,rgb,lsb,xy    .. file: OpenPGP Secret Key
b2,bgr,msb,xy    .. file: OpenPGP Public Key

(kali@kali)-[~/Downloads/forensics_ctflearn/zz]
$
```

zsteg

Searched alot about OpenPGP. By alot I mean ALOT. Turned out to be noise, not important. Read as well on the whole "67" approach which turned out to be some gen Z stuff, considred as well to be brainrot . Its even there as a meme in some games.



Clash Royale

mmm. Lets dive deep then, wont sleep till I solve it.

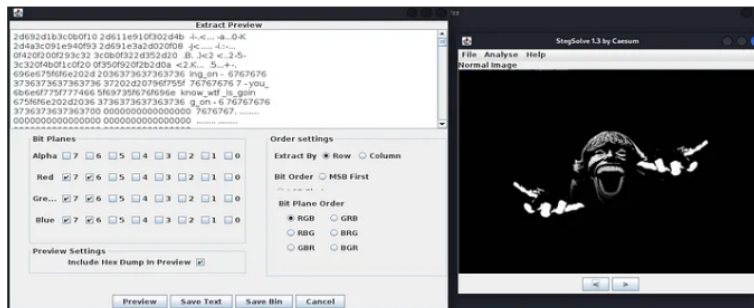
I got an idea, the image is in grey scale, which means it is actually 8 bit, which means there are 8 bit planes from 0 to 7. Make sense?

```
(kali@kali)-[~/Downloads/forensics_ctflearn/zz]
$ java -jar stegsolve.jar
Picked up _JAVA_OPTIONS: -Dawt.useSystemAAFontSettings=on -Dswing.aatext=true
```

stegsolve

By diving deep i mean lets analyze the image's individual bit planes and color channels. Maybe even applying various transformations.

Lets start first by trying to extract data from the image, of course using LSB as the bit order. Why? remeber the colorful pixels?



67 (literally) which was a hint for the planes

After some hours of thinking and searching, and I think we found a crucial hint. but is it? do i really "know_wtf_is_goin"? Not even a bit!

This is a mess, but its a main aspect in the recipe.

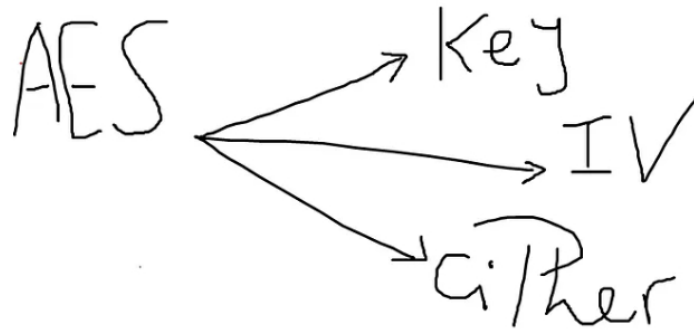
At this point .. lets return back to the first hint.

hint indicating AES and some math

Damn. This is going for the AES!



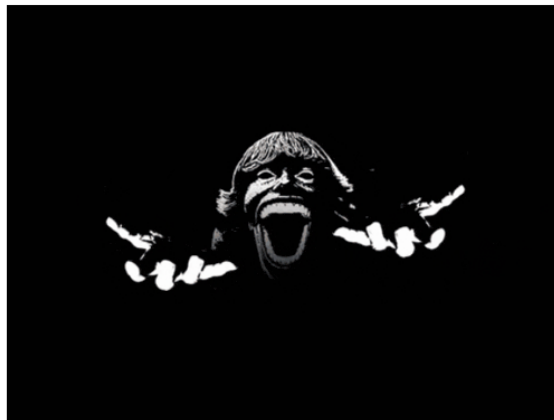
But what does the math mean? 2 halves which gives me the full thing??



AES

Didn't get it at first. Did a lot of trial and error on CyberChef as well but useless. Here my brain started to get smoked, again! xD

CTFs need patience.



Official GIF of the challenge xD

Lets try again. We just need to get the right cipher text, in addition to the right intialization vector and the key. Which we already extracted using StegSolve.

I just want you to know that IV is actually initialization vector, which is 128 bits. Here we are dealing with hex, so we need:

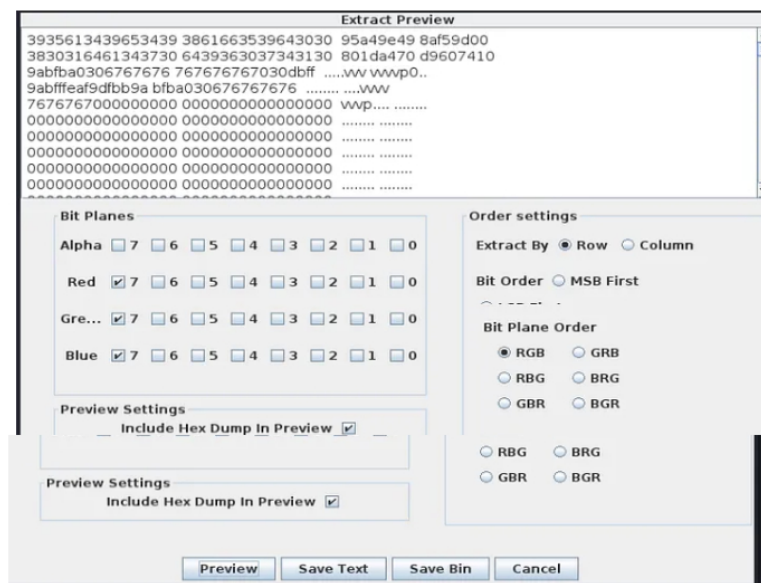
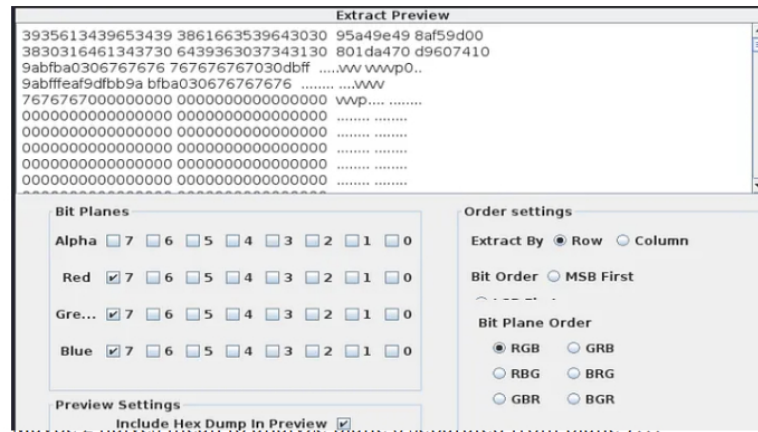
$$32 \text{ hex characters} \times 4 \text{ bits per hex digit} = 128 \text{ bits}$$

Here's how it breaks down:

- 1 byte = 8 bits
- 1 byte = 2 hex characters
- AES IV = 16 bytes = $16 \times 8 = 128$ bits
- 16 bytes = 32 hex characters

Here I knew I had the key already and the IV. The whole problem was in the cipher text, that's why my trials failed. Remember the the math hint, and the whole 67 thing? Think Mousa, Think!!!

Maybe 2 halves mean to analyze plane 6 separated from plane 7???



bit plane 6

Now we are talking. Now we have total 64 bits of ciphertext.



So now, we have the following:

Initialization Vec: 36373637363736373637363736373637 (16 BYTES {1 ROW})

CIPHER: 64 bytes (DONE)

Key: guess what? xD (most important its 24 bytes)

AES Size Rules (What's Fixed vs. What Can Grow)			
Component	Fixed or Variable?	Size	Notes
Key	Fixed	128 / 192 / 256 bits (16 / 24 / 32 bytes)	Chosen by the AES variant (AES-128, AES-192, AES-256).
IV (Initialization Vector)	Fixed	128 bits (16 bytes)	Always 128 bits for AES (only used in certain modes).
Block size	Fixed	128 bits (16 bytes)	AES always encrypts in 128-bit blocks internally.
Ciphertext	Variable	Depends on plaintext length	The longer the plaintext, the longer the ciphertext (multiples of 16 bytes in block modes).

Recipe

To Hex

Delimiter: Space, Bytes per line: 0

Input: you_know_utf_is_going_on

Output: 79 6f 75 5f 6b 6e 6f 77 5f 77 74 66 5f 69 73 5f 67 6f 69 6e 67 5f 6f 6e

24 bytes ✓

Baking the key

Recipe

AES Decrypt

Key: 796f755f6b6e6f775f7774665f69735... HEX

Input: fcc0d84ccb89f7a2146ec5dde324497c95a49e498af59d00081da470d9607410

Output: 79 6f 75 5f 6b 6e 6f 77 5f 77 74 66 5f 69 73 5f 67 6f 69 6e 67 5f 6f 6e

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Recipe

To Hex

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Bytes per line: 0

Input

you_know_utf_is_going_on

24 bytes ✓

Output

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To Hex

Delimiter: Space

Bytes per line: 0

Input

you_know_utf_is_going_on

1 1 1 ✓

[Redacted]

[Redacted]

[Redacted]

[Redacted]

[Redacted]

[Redacted]

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